

First hackathon

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Project **SEE BEYOND THE PIXELS** Technical Report

1. Executive Summary

Project Vortex is a sophisticated AI-generated image detection system designed to distinguish between authentic (Real) media and synthetic (AI) content. The system utilizes an Ensemble Learning approach, combining multiple specialized deep learning models with heuristic analysis (Frequency & Identity Risk) to achieve high accuracy. The core innovation is "Gap Trap V3," a logic layer that resolves conflicting signals in the "uncanny valley" of detection confidence.

2. **Model Architecture & Core Logic**

2.1 Ensemble Models

The system does not rely on a single model. Instead, it aggregates predictions from distinct neural networks:

1. General Detector

(dima806/ai_vs_real_image_detection):

- Architecture: Transformer-based (Hugging Face AutoModelForImageClassification).
- Role: Serves as the primary classifier. It detects general synthetic patterns, artifacts, and texture inconsistencies common in AI generation (Midjourney, DALL-E, etc.).
- Behavior: Acts as a "Conservative" baseline.

2. Specialized Deepfake Detector

(prithivMLmods/Deep-Fake-Detector-v2-Model):

- Architecture: Likely a fine-tuned Vision Transformer (ViT) or specialized CNN.
- Role: Focuses specifically on facial manipulation and deepfake artifacts.

- Behavior: More sensitive to facial inconsistencies.

3. Vision Transformer Training Pipeline:

- The project includes a training infrastructure (

train.py) utilizing Google's ViT (Vision Transformer) vit-base-patch16-224.

- This pipeline allows for fine-tuning custom models on specific datasets (FaceForensics+, custom scrapes) to further specialize detection capabilities.
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2.2 "Gap Trap V3" Logic

To eliminate false positives (real images marked as AI) and false negatives (AI marked as Real), the system implements a custom decision engine:

- Zone A (Filtered Real Override): If the Face Model is extremely confident (>75% Real), the system overrides a weak "AI" signal from the

General Model, attributing it to beauty filters rather than generative AI.

- Zone B (The Trap): If the Face Model is indecisive (0.25 - 0.65), and the General Model suspects AI, the system enforces the "AI" verdict. This captures "Uncanny Valley" images—high-quality fakes that don't have obvious artifacts but lack true biological texture.
- Zone C (High Quality Real): If facial textures score highly for realism (>0.65), the system prioritizes this signal over general noise patterns.

3. Advanced Analysis Modules

3.1 Frequency Analysis (Signal Processing)

AI-generated images often leave distinct fingerprints in the frequency domain that are invisible to the naked eye.

- FFT Spectrum: Uses Fast Fourier Transform (2D FFT) to convert the image from the spatial domain to the frequency domain. Regular

geometric patterns (stars/spikes) in the spectrum often indicate upsampling artifacts common in GANs and Diffusion models.

- Pattern Map: Applies a Laplacian High-Pass Filter to isolate high-frequency noise. This visualizes the "checkerboard" artifacts often left by deconvolution layers in neural networks.

3.2 Identity Risk Detection (Heuristics)

A dedicated module for analyzing the suitability of an image for ID verification, minimizing identity theft risk.

- Technology: OpenCV (cv2) with Haar Cascade Classifiers.
- Analysis Criteria:
 - Face Count: Ensures exactly one face is present.

- Alignment: Checks eye coordinates to ensure head roll is < 10 degrees.
- Centering: Verifies the face is within 15% deviation from the image center.
- Quality: Uses Laplacian Variance (>50) to detect blur.
- Background: Analyzes the variance of the top pixel strip to ensure a plain, uniform background.

4. Technical Stack

- Backend Framework: Flask (Python)
- Deep Learning Engine: PyTorch & Hugging Face Transformers
- Image Processing: Pillow (PIL), NumPy, OpenCV
- Frontend: HTML5, CSS3, JavaScript (Dynamic UI)